

Interpretation of Results

1. **Summarize your findings by interpreting their physical/engineering meaning (for example, why would engineers want to compute the analytical results in Task 3?):**

TASK 1:

- Step 1 (Alex) - The example lithium-ion battery dataset was loaded into MATLAB, and several charging cycles were randomly selected from different points throughout the dataset to observe how the battery's charging behavior changed over its lifetime. Since the dataset contains hundreds of charging cycles, sampling cycles from the beginning, middle, and end provided a representative overview without having to analyze every cycle individually. By comparing the voltage and current curves, it became clear that the battery's charging profile degraded as it aged. The user of the livescript is then prompted to make a cycle number selection after seeing 12 example charging curves interspersed through the dataset. Given the degradation of charging cycles, we recommend Cycle 1 as a selection, but chose to still make the cycle selection option to keep it interactive (this input prompt appears in the Command Window). Cycle 1 is recommended because it had the smoothest voltage and current curves with the fewest discontinuities, making it the best candidate for fitting a simple RC charging model.
- Step 2 (Alex) - The battery data was preprocessed to isolate only the charging portion of Cycle 1 (later changed to apply to any user-selected cycle number). Since the goal was to model the charging profile, removing the discharging portion ensured that the analytical model was fit only to the relevant data. This preprocessing step improves the accuracy of the curve fit and allows the charging behavior to be analyzed without interference from the discharge cycle.
- Step 3 (Alex) - The charging voltage data was fit to a simple RC capacitor charging equation using MATLAB's curve fitting tools. Since the battery's maximum voltage of 3.6 V was already known, the only unknown parameter was the RC time constant, τ . This value represents how quickly the battery approaches its maximum charging voltage. Engineers estimate this parameter because it provides a simple mathematical model of the battery's charging behavior, which can be used to predict charging performance, compare batteries, and develop battery management system (BMS) algorithms.
- Step 4 (Alex) - The measured charging data and fitted RC model were plotted on the same graph to visually compare how well the analytical equation matched the experimental battery data. The fitted curve closely followed the overall charging trend, showing that a first-order RC model provides a reasonable approximation of the battery's

charging profile, even though real lithium-ion batteries exhibit more complex electrochemical behavior.

- Step 5 (Alex) - The goodness-of-fit errors, including the sum of squared errors (SSE), root mean squared error (RMSE), and coefficient of determination (R^2), were calculated to evaluate how well the RC model matched the measured charging data. These values provide a quantitative measure of the model's accuracy, with lower SSE and RMSE values and an R^2 value closer to 1 indicating a better fit. Engineers use these errors to determine whether a simplified analytical model is accurate enough for predicting battery behavior, comparing different battery models, and supporting the development of battery management systems.

TASK 2:

- Step 1 (Chris) - The Voltage vs Time graph shows how the battery's terminal voltage increases throughout the charging cycle. The voltage initially has a steep incline (for cycle 1) during the CC phase, then steadily tapers as it reaches the battery's maximum charging voltage (in the CV phase). Engineers analyze this voltage profile as it provides insight to the battery's charging patterns as well as the health. The shape of the curve that is created is a basis for several of the analytical calculations prevalent in task 3, such as the rate of voltage charge, and the duration of time required to reach certain percentages of charge. These calculations help engineers improve the efficiency of the battery.
- Step 2 (Chris) - The Current vs Time graph depicts how charging current changes throughout the charging process of the battery. In cycle 1, the battery's current remains constant during the CC phase, as to transfer energy into the battery, and once the battery reaches max voltage, it switches to the CV phase until the charging is complete. The analysis of current behavior is important as it determines the amount of heat generated in the battery, which is important information to help engineers improve the efficiency of the battery, minimize the amount of heat loss, and improve the lifespan of the battery.
- Step 3 (Vincenzo) - The power vs time graph displays the rate of energy transfer throughout one of the battery's charging cycles. Engineers are able to analyze this data in order to identify periods of high energy transfer, any unusual discrepancies or gaps, and evaluate the efficiency of the charging processes. We can also compute the total amount of energy transferred by integrating the area under the curve, which is done during Step 3 of Task 3.

TASK 3:

- Step 1 (Vincenzo) - The numerical derivative of voltage with respect to time was calculated in this section, which mainly allows engineers to visually observe the battery's charging characteristics after graphing the data. Instead of looking at a voltage vs time

graph, the rate-of-change data provides much more insight on how rapidly the voltage is changing throughout the entire cycle. The minimum and maximum points help us identify important phases of the charging process and will be useful when trying to improve the overall efficiency of the battery's charging.

- Step 2 - Determining the time needed to reach 80% and 100% charge essentially tells us how fast the battery can charge to both an acceptable amount, and its maximum capacity. These benchmarks help engineers compare charging methods while also maintaining healthy battery life for long term use. Too slow of a charge is undesirable while too fast of a charge can lead to increased battery decay and internal stress, so charging times are very significant pieces of data to pay close attention to.
- Step 3 - Calculating the total energy delivered to the battery during the charging cycle gives us valuable insight on how much charge is being stored in the battery at any given point. Since this data is taken during the battery's charging phase, it should be apparent that the total energy (J) is positive. This metric is useful because it allows engineers to evaluate the battery's performance, estimate costs, and determine how efficiently the battery can hold its charge. Undesirable discharge and current leakage are common problems for batteries, so it's vital to understand the properties of the battery.
- Step 4 (Chris) - The calculations of resistive energy loss helps estimate how much energy is converted into heat due to the battery's internal resistance while it is charging. Because the current flows through the battery's internal resistance, some energy is dissipated instead of stored. This value is important to compute to determine the amount of energy waster, to optimize charging, increase battery lifespan, and improve the efficiency of the battery. These calculations can also help to reduce the amount of heat generated during charging.

2. What are some limitations of your work?

Some limitations of the work include experience with MATLAB coding, due to inexperience in current levels of programming leading to primitive coding techniques. Another limitation includes time and responsibilities. Although we were given a specific amount of time to complete our project, other responsibilities outside of this internship, including school, jobs, and extracurriculars, limit the amount of time we are able to dedicate to completing the advanced portions. Lastly, our analysis is only based on the data given to us, for example voltage, current, temperature, internal resistance. Thus, the results may not fully represent the true nature of the battery under different environments.

3. What are some practical next steps?

Practical next steps for this project would be to determine and plot the rate of change of the current/power with respect to time, as that was an analysis required for voltage. This analysis could provide greater insight about how rapidly the charging conditions can change during each phase. Another step could be to determine what factors could have an effect on the internal resistance of the battery, such as temperature, or testing different charging profiles with the use of Simscape battery.